



Outline of the lecture

The history of Informatics & HCI

Our roots and evolution

Conceptual Foundations

Information Processing | Gestalt Theory | Functional Approaches

Action Research & Praxeology

Work(place) studies & ethnography | Contextual Approaches |
Praxeology & Appropriation | Activity Theory

Structuration & Infrastructuring

A holistic look at Human-System-Interaction

HCI in Practice

Designing | Prototyping | Evaluation

Users and Use Context

Basics

Primary Users	Secondary Users	Tertiary Users
Those who directly interact with the technology	Those who use it via an intermediary; may have an influence about primary users (using/not using it)	Those who are affected by the technology use and decide its purchase/implementation

Use Context

- Where are users located?
- What are the social, cultural and organisational aspects of the context?
- With what/whom do they interact?
- What role do technologies play in the context?
- How do these aspects impact upon the way that users do things?

stakeholders

motivations, and goals

understand people's behaviors

An individual or group with an interest in the success of a product/organization in delivering intended results and maintaining the viability of the organization's products and services. **Stakeholders influence** programs, products, and services (Kotonya and Sommerville, 1998)

Before Use Context

Problems and deficits in HCI in early 1980

Problem situation

HCI concentrates on users and interface. However, interaction with other people, context, culture, groups is rather neglected (Norman, 1980).

Observable deficits in cognition-based HCI*

1. Early advanced user interfaces assumed the **users mental models or generic users**, their cognition - without concern for qualifications, work environment, etc.
2. The **role of the artifact** as it stands between the user and the materials, objects and outcomes was ill understood.
3. In validating findings and designs there was a heavy focus on novice users whereas everyday use by experienced users and concerns for the development of **expertise were hardly addressed**.
4. Detailed task analysis **failed to capture the complexity of real-life action**
5. From the point of view of complex work settings, it was striking how most **HCI focused on one user – one computer** in contrast to the ever-ongoing cooperation and coordination of real work.
6. Users were mainly seen **as objects of study**.

*Bertelsen, Olav W., and Susanne Bødker. "Activity theory." *HCI models, theories, and frameworks: Toward a multidisciplinary science* (2003): 291-324.

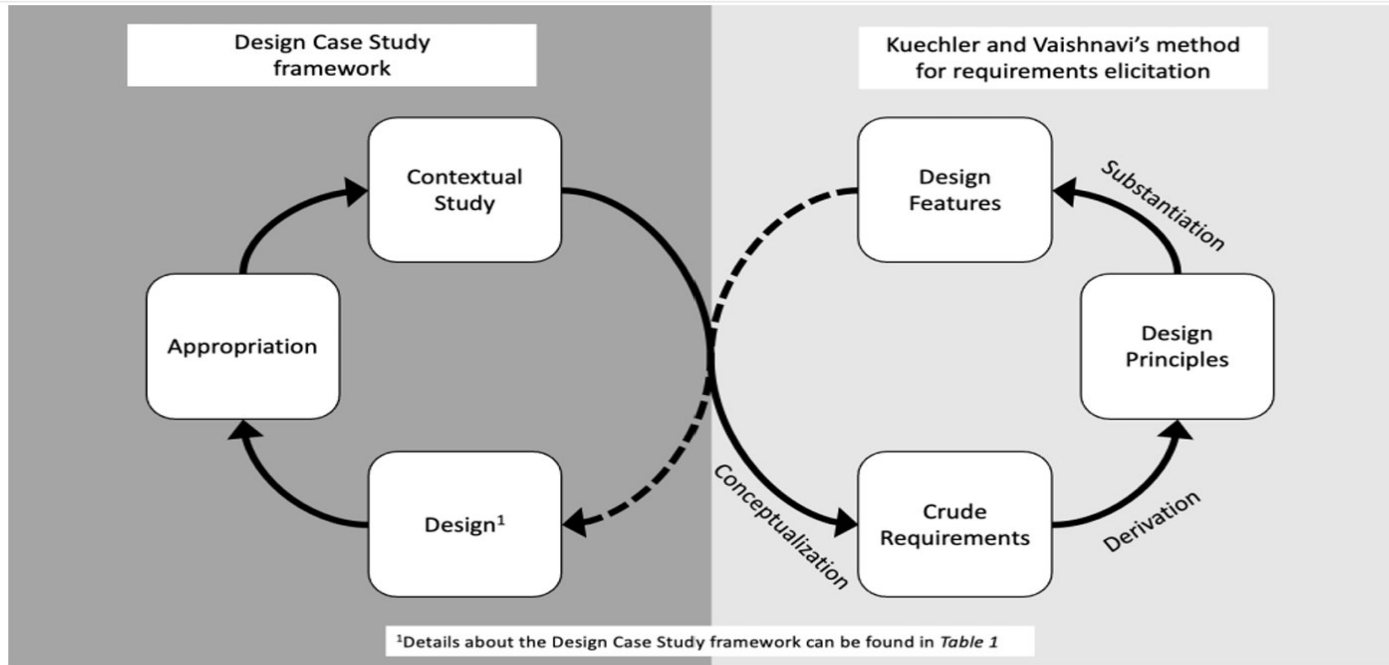


Figure 4. Blending adaptation of Kuechler and Vaishnavi's for requirements elicitation within stages of DCS.

Pragmatic Approach

Practice Centered Design / Design Case Studies

Formative and Contextual Approaches

Not a clear difference

	Formative	Contextual
Purpose	Shape and improve design	Understand real-world context
Stage	During design (iterative)	Before or during early design
Outcome	Inform design, Design insights and refinements	Contextual understanding and user requirements

Contextual Approaches

Basics

Approach	Description	Type of Data Collected
Contextual Inquiry	A semi-structured interview conducted in the user's real work or use environment; the researcher observes and asks questions as the user performs tasks.	Task flows, workarounds, artifacts used, implicit practices
Ethnographic Fieldwork	Long-term observation of users in their natural environment; researcher becomes part of the community or workflow.	Social interactions, routines, culture, communication patterns
Participant Observation	Researcher participates in the user's activity while observing interactions.	Detailed, experiential understanding of behaviors and meanings
Diary Studies / Experience Sampling	Participants record activities, feelings, or interactions with a system over time.	Longitudinal data on user experiences, habits, frustrations
Cultural Probes	Creative, open-ended prompts (e.g., postcards, disposable cameras, diaries) given to users to express their experiences.	Subjective and emotional insights into user values and contexts
Shadowing / Field Studies	Researcher shadows a participant during tasks to observe use in context.	Workflow and contextual factors (e.g., interruptions, collaboration)

Formative and Contextual Approaches

Basics

Formative Method	Description / Purpose	Type of Data	When Used in Design Process
Interviews (Semi-structured or Unstructured)	Explore users' experiences, needs, motivations, and pain points in their own words.	Qualitative	Context study
Focus Groups	Group discussion to explore attitudes, preferences, or reactions to ideas or prototypes.	Qualitative	Context to mid design
Participatory Design Workshops	Involve users directly in co-designing solutions or interfaces.	Qualitative / Design artifacts	Mid design
Card Sorting	Explore how users group and label information for navigation or categorization tasks.	Qualitative / Quantitative	Early or mid design
Storyboarding / Scenario-Based Design	Develop stories or visual scenarios of how users might use a system.	Qualitative / Design artifact	Early or mid design
Prototyping (Low- to Mid-Fidelity)	Create simplified models of the design to test ideas early.	Qualitative / Mixed	Mid design (iterative)
Heuristic Walkthroughs (Formative Usability Evaluation)	Designers or experts identify usability issues early, before full testing.	Qualitative	Mid design
Field Studies / Field Trials	Observe how users interact with prototypes in real settings.	Qualitative / Mixed	Mid to late design

Intensifying the Problem: Work(place) studies

Basic idea: HCI takes place against the background of a work situation (everyday practice).

▶ **Happening in the wild (cannot be Lab Studies)**

Cognitive approaches fail to capture the complexities and nuances of real-world environments where work actually happens - involving unexpected events, evolving goals, and constant adaptation - so interruptions, social interactions, environmental constraints cannot be adequately modeled

▶ **Is Messy**

Real work is messy, non linear and collaborative - requiring iterative, overlapping, and occasionally conflicting actions

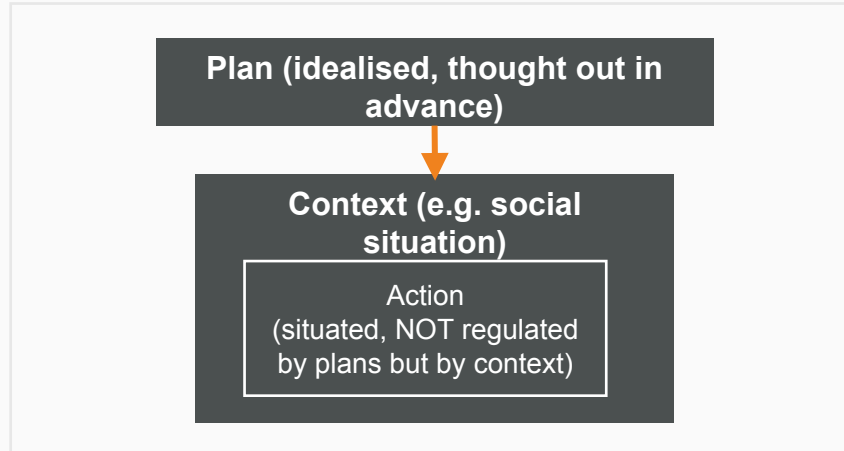
▶ **Turn to practice**

By focusing on actual practices, researchers can uncover tacit, context-specific knowledge that is invisible in purely cognitive approaches, which prioritize universal theories over situated understanding



Plans & Situated Action

Ethnographical foundation for Work(place) studies: a subjective orientation to action



Pioneer: Lucy Suchman, 1987

Planning and representation, central to symbolic theories, are claimed to be irrelevant in everyday human activity (Situated action: A symbolic interpretation by Vera and Simon, 1993)



Cognitive Models (esp. from psychology)

People's behaviour driven by intellectual models, purpose, aim, motivation etc. Explained by theories.



Situated Action

Humans and their interactions with the world cannot be understood using symbol-system models and methodology, but only by observing them within real-world contexts or building nonsymbolic models of them.



Ethnographical approach

Practical orientation: "What is happening here?" vs. "which theories correspond to what is happening?".

Origins lie in the research of indigenous peoples "from the inside" –participation in the field is an integral part of this method! Its principal direction is ethnomethodology: "*We cannot recognize one objective, actual practice but rather the actor's intended practice*".

"The world is changing rapidly, with new views on the role of design in the world as well as a growing realization that the existing model of design practice had its historical origins in the industrialized nations of the world as a service to industry. Today's designers have moved far beyond these historical origins. One result is greater awareness of the deterioration of the economies, environment, and culture in what has been called the "colonization of the world."

Meyer, M. W., & Norman, D. (2020). Changing Design Education for the 21st Century. *She Ji: The Journal of Design, Economics, and Innovation*, 6(1), 13-49. Changing Design Education for the 21st Century - Science Direct United Nations. (2019). (Source: Science Direct)

Role of a Designer

Activity Theory (AT): one analytical artifact

History / Why AT in early 1990

History of AT

The fathers of AT were Russian Psychologists Vygotsky (1962) and Leontiev (1978) – a framework to understand ***mental functions as developing from the performance of external actions***

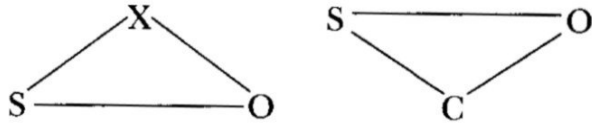
In 1987, discussions and use of AT in HCI were primarily framed within the Scandinavian activity theory strand, developed by Yrjö Engeström, to satisfy a need, ***actions only make sense in a social context of a shared work activity between subject and object***

The goal of AT is understanding the mental capabilities of a single individual. However, it rejects the isolated individuals as insufficient unit of analysis, analyzing the cultural and technical aspects of human actions. AT is most often used to describe actions in a **socio-technical system** through six related elements.

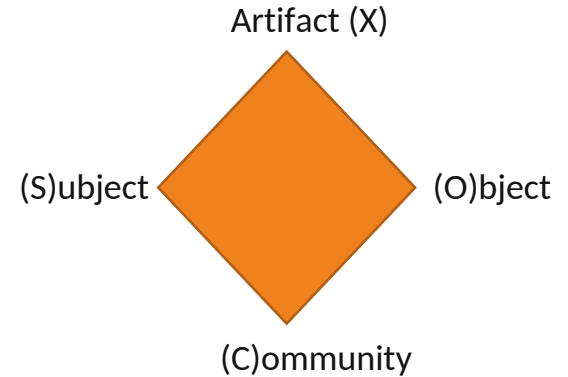
Transfer to IT: *Developmental Work Research* (Engeström 1987, Kuutti and Arvonen 1992) / *Through the interface*: Bodker 1991 / Overview in Nardi 1996

Activity Theory (AT)

How to read the AT



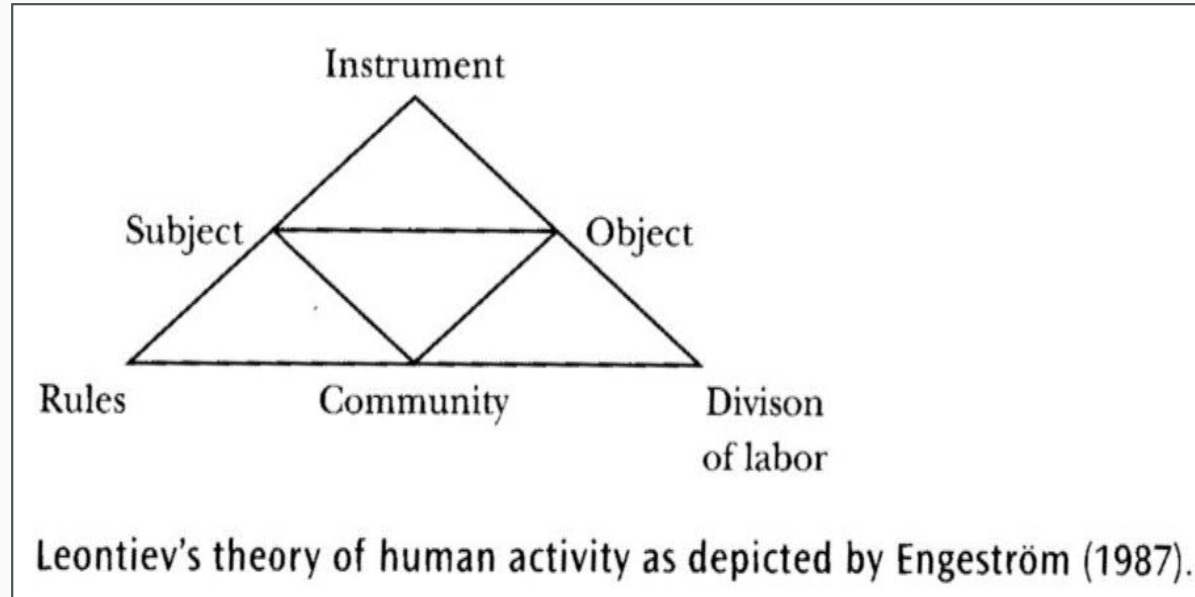
Triangles of activity. On the left is human activity mediated by artifacts (Vygotsky); on the right socially mediated activity (Leontiev).



Bertelsen, Olav W., and Susanne Bødker. "Activity theory." *HCI models, theories, and frameworks: Toward a multidisciplinary science* (2003): 291-324.

Activity Theory (AT)

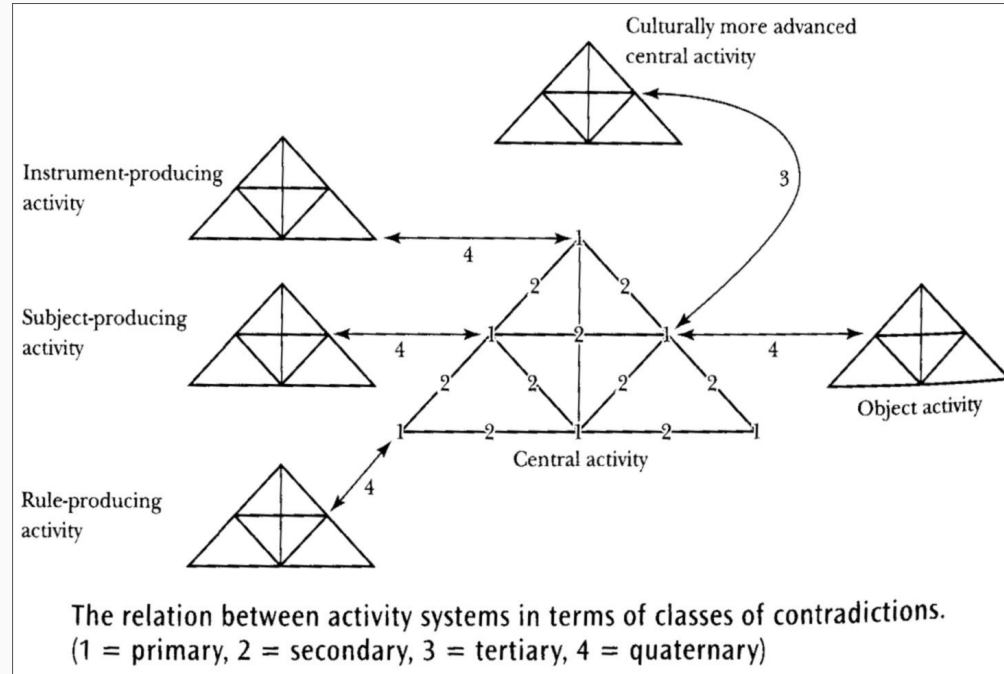
How to read the AT



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Activity Network (AT)

How to read the AT

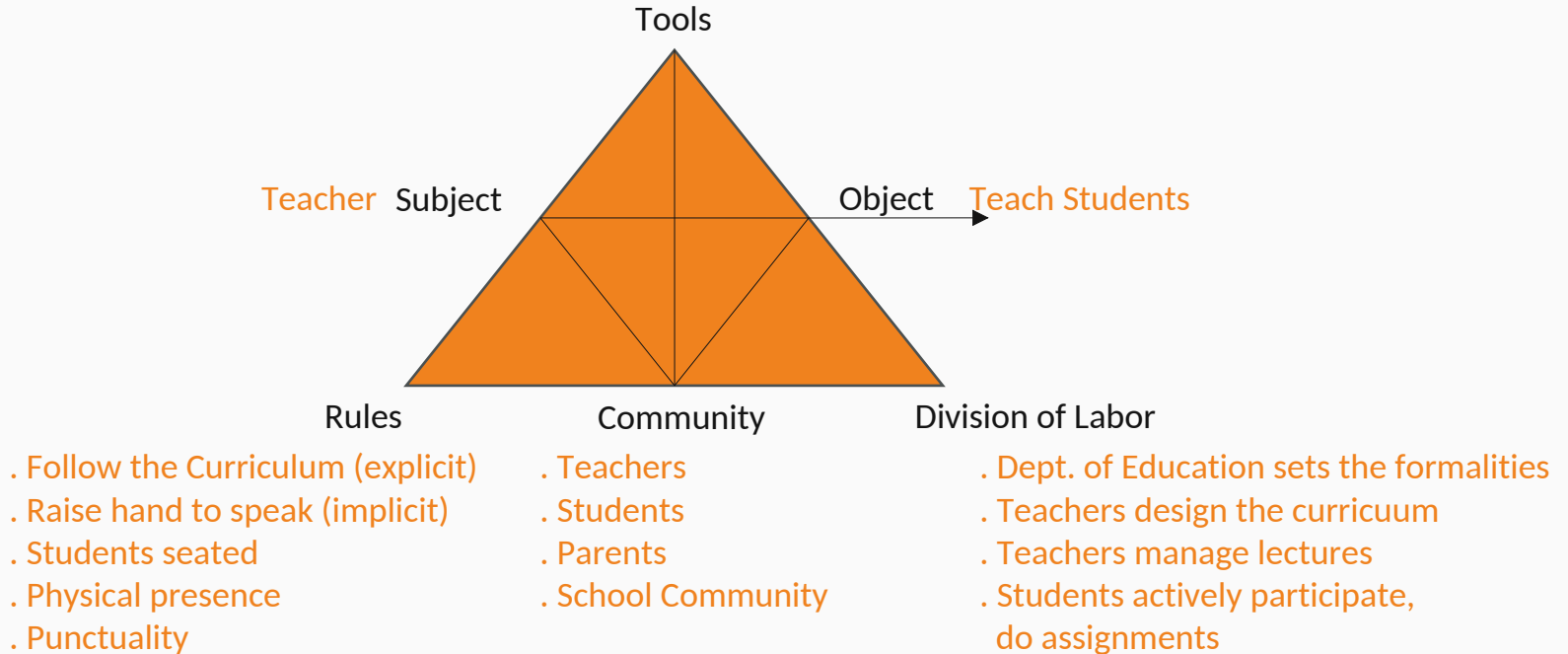


Bertelsen, Olav W., and Susanne Bødker. "Activity theory." *HCI models, theories, and frameworks: Toward a multidisciplinary science* (2003): 291-324.

Activity Theory (AT)

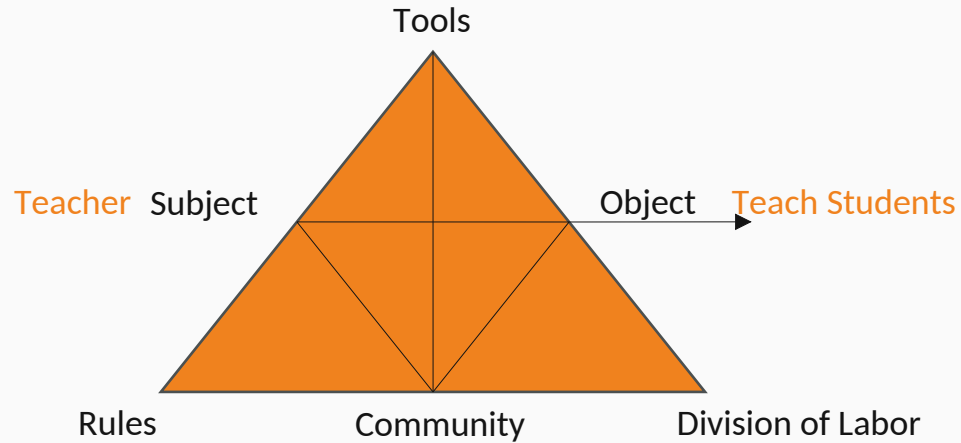
Example: Physical Classroom

- . Eye Contact
- . Body Language
- . Blackboard
- . Chalk
- . Textbooks



Activity Theory (AT)

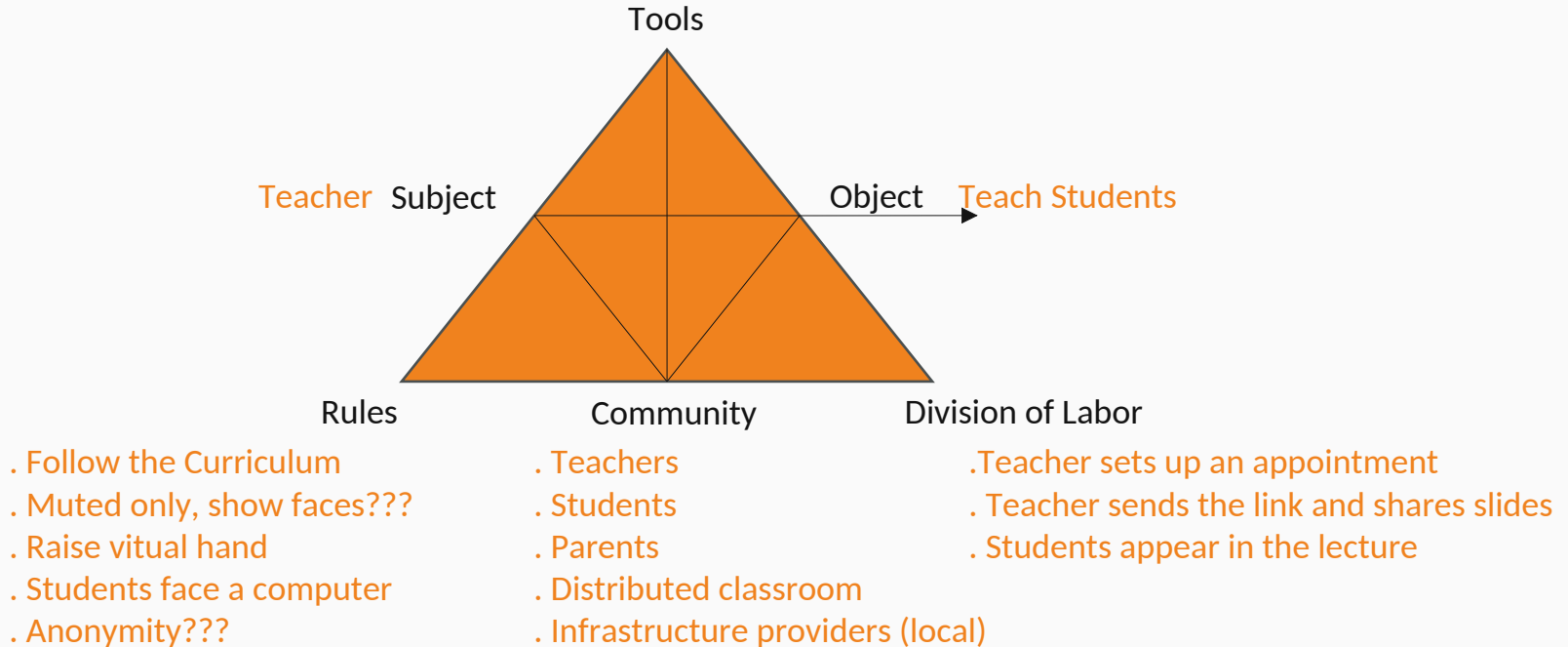
Example: Virtual Classroom



Activity Theory (AT)

Examples: Virtual Classroom

- . Voice
- . Email
- . Text messaging
- . Zoom
- . Breakout rooms



AT: Application possibilities in HCI

Case example: Check lists

Check lists (Korpela et al. 2000)

Activity related questions:

- Outcome: Which product/services?
- Object/Process: Raw materials and their contribution to the product?
- Instruments: Which tools, abilities, knowledge?
- Subjects: Who is doing what, exactly?
- Social relationships/means: Which conventions, rules, division of labour, communication rituals?

Activity network related questions:

- Outcome: Who needs the product/service, and what for?
- Object/Process: Where do the raw materials and their contribution to the product originate?
- Instruments: Where do tools, abilities and knowledge come from and how are they generated?
- Subjects: How do subjects obtain their useful abilities?
- Social relationships/means: How are conventions, rules, division of labour and communication rituals “produced”?

Activity Theory Sample Qs

Individually, build the AT of each context

CULTURAL

NGO supports
women who
suffers
domestic
violence

ECONOMIC

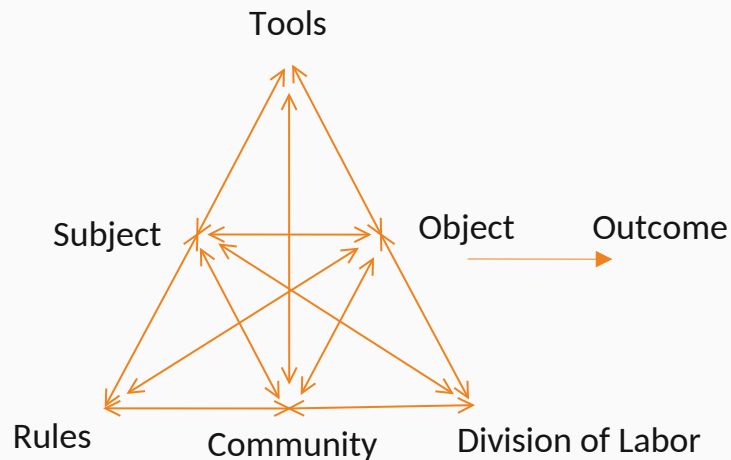
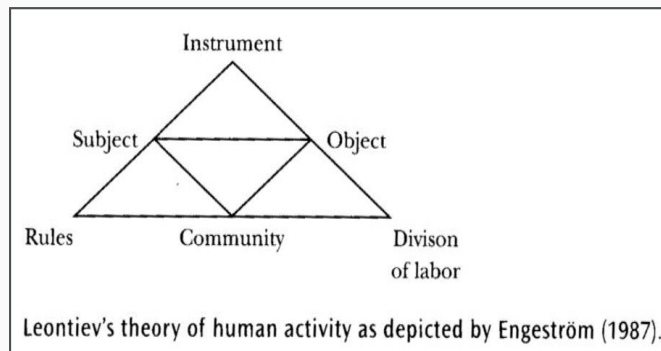
NGO offers job for migrants

TECHNICAL

Public services
to **citizens** with
limited access
to the internet

SOCIAL

Public services
to **citizens** who
are illiterate



Activity Theory

Case Study

Climate activists clashed with German police in a continuing standoff over the demolition of this German village (Lützerath) for the expansion of a coal mine. Activists who had occupied Lützerath planned, in partnership with an NGO, to build a system to inform them about the approach of the police and mining machinery. Their intent is to protect the 1.5 degree-limit, as set out in the Paris Climate Agreement, by blocking the open coal mine. Through Participatory Design, an app with acoustic monitoring systems was developed to gather real-time data from the region. Sensors and microphones with solar panels were installed on trees located in critical areas. In case of an uncommon disturbance such as the noise of a police siren or the machinery from the mining, real-time alerts would be sent to the local activists and the activists around the world with the geo-localization of that sound. Immediately, they could get together to start a demonstration which can result in halting the machinery, while enabling biodiversity measurement and monitoring.

Activity theory (AT) has shown much promise as a framework for Human Computer Interaction, particularly for analyzing complex activity and its context. AT is an answer to the problem of studying isolated individuals in the laboratory setting, instead of dealing with the isolated relation between the subject (S) and an object (O). Vygotsky (1962) and Leontiev (1978) introduced the factors of mediating tools (X) and communities (C). 2- Based on the case study above, who or what could be the S, O, X and C factors? Can you represent activity theory in a graph/drawing, like Engeström (1987) did, including the new factors along with the four factors mentioned above?

AT: Goals / Advantages

6 central advantages



Inclusion of a historical perspective



Analysis and design of a work practice under consideration of qualifications, work environment, division of labour etc.



Analysis and design under consideration of actual usages and complex user interdependencies: inclusion of artefacts as a mediator



Inclusion of the development of user knowledge and usage in general



Activation of users in the design process



There is no need to restrict studies to the level of 'general users'; specific practical background can be addressed

AT: Application possibilities in HCI

Case example I

Further development of activity networks as a design goal

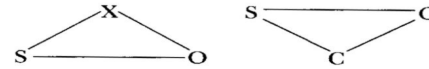
- Context menus/“Direct manipulation“ to describe the ‘**zone of proximal development**’ (also Bardram and Bertelsen 1995)

The shaping of user interfaces (Beaudouin-Lafon 2000)

- The understanding of screen elements (Drop-down-Menu, scrollbar, radar navigation etc.) as artefacts for the manipulation of objects (e.g. text documents)
- The understanding of input devices (mouse, keyboard, joystick etc.) as artefacts for the manipulation of screen elements

The design of software systems (Susanne Bodker)

- The development, supported by theory, of styles of user interface
- Theory-supported check lists for requirement analysis



Triangles of activity. On the left is human activity mediated by artifacts (Vygotsky); on the right socially mediated activity (Leontiev).

AT: Summary

Advantages, scope and limitations

AT focuses on artefacts

Instruments and products with which agents interact.
Therefore easily empirically ascertainable (!?)

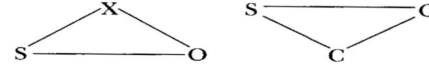
AT makes it possible to see and assess the role of computer applications as tools for action :

To avoid technical tunnel vision, the focus is on agents, actions and artefacts: reflections remains (relatively) neutral regarding work equipment and results
Instruments, their embedded action culture and their history of supersession open up historical perspectives action systems

AT is also a cultural learning theory

Consideration of development steps against a historical background

AT initially only a mental framework which must be completed methodologically: Good analytical methodological basis. Implementation of AT perspectives in design is not easy. Pragmatism? Design orientation?



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